

Vaultwarden Security Audit Report

Part 4: Informational Findings & Security Recommendations

Audit Date: January 2026 **Application:** Vaultwarden v1.0.0 **Auditor:** Security Assessment Team **Repository:** <https://github.com/dani-garcia/vaultwarden>

Executive Summary

This final report documents **8 Informational findings** that represent security best practices, hardening opportunities, and operational improvements. While these findings do not represent direct vulnerabilities, they provide valuable guidance for enhancing Vaultwarden's security posture, operational resilience, and compliance readiness.

Report Structure

Part I: Informational Findings (8) - Security configuration opportunities - Monitoring and observability gaps - Dependency management improvements - Documentation enhancements

Part II: Comprehensive Security Recommendations - Implementation priority matrix - SOC 2 compliance roadmap - Security testing strategy - Deployment hardening guide

Part I: Informational Findings

I-1: No Automated Dependency Vulnerability Scanning in CI/CD

Severity: Informational **Category:** Vulnerability Management **SOC 2 Relevance:** Yes (CC7.1 - System Operations)

Description

While Vaultwarden has Trivy scanning enabled in CI/CD for container image vulnerabilities, there is no automated Rust dependency vulnerability scanning using tools like `cargo-audit` or

`cargo-deny`. This means that vulnerable dependencies could be introduced through normal development without immediate detection.

Current State

Trivy Scanning Present:

File: `.github/workflows/trivy.yml`

```
- name: Run Trivy vulnerability scanner
  uses: aquasecurity/trivy-action@b6643a29fec7f34b3597bc6acb0a98b03d33ff8
  with:
    scan-type: repo
    ignore-unfixed: true
    format: sarif
    output: trivy-results.sarif
    severity: CRITICAL,HIGH
```

Missing: Rust-specific dependency auditing

No `cargo-audit` or `cargo-deny` in CI/CD pipeline - Manual dependency review only - No automated vulnerability database checks - No policy enforcement for dependency licenses or sources

Risk Analysis

Impact: - Vulnerable dependencies may be merged without detection - No automated notification of newly discovered CVEs in existing dependencies - Delayed response to dependency vulnerabilities - Compliance gaps (SOC 2 requires vulnerability management)

Likelihood: Medium (new vulnerabilities are discovered regularly)

Real-world Examples: - CVE-2020-35711: Rust crypto library vulnerability (fixed in later versions) - CVE-2021-45707: tokio vulnerability requiring version update - Supply chain attacks on popular Rust crates (rare but increasing)

Recommendations

Priority: Medium

1. Add cargo-audit to CI/CD:

```
# .github/workflows/security-audit.yml
name: Security Audit

on:
  push:
    branches: [main]
  pull_request:
    branches: [main]
  schedule:
    - cron: '0 0 * * *' # Daily at midnight

jobs:
  cargo-audit:
    name: Cargo Audit
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Install cargo-audit
        run: cargo install cargo-audit

      - name: Run cargo audit
        run: cargo audit --deny warnings
        continue-on-error: false

      - name: Upload audit results
        if: failure()
        uses: actions/upload-artifact@v4
        with:
          name: cargo-audit-report
          path: target/audit-report.json
```

2. Add cargo-deny for policy enforcement:

```
# .github/workflows/cargo-deny.yml
name: Cargo Deny

on: [push, pull_request]

jobs:
  cargo-deny:
    name: Cargo Deny
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Cargo Deny
        uses: EmbarkStudios/cargo-deny-action@v2
        with:
          log-level: warn
          command: check
          arguments: --all-features
```

3. Create deny.toml configuration:

```
# deny.toml
[advisories]
db-path = "~/.cargo/advisory-db"
db-urls = ["https://github.com/rustsec/advisory-db"]
vulnerability = "deny"
unmaintained = "warn"
yanked = "deny"
notice = "warn"
ignore = []

[licenses]
unlicensed = "deny"
allow = [
    "MIT",
    "Apache-2.0",
    "BSD-3-Clause",
    "ISC",
    "MPL-2.0",
]
deny = [
    "GPL-3.0",
    "AGPL-3.0",
]

[bans]
multiple-versions = "warn"
wildcards = "allow"
highlight = "all"

[sources]
unknown-registry = "deny"
unknown-git = "deny"
allow-registry = ["https://github.com/rust-lang/crates.io-index"]
```

4. Local development integration:

```
# Add to development documentation
$ cargo install cargo-audit cargo-deny

# Run before commits
$ cargo audit
$ cargo deny check
```

Benefits: - Automated CVE detection for Rust dependencies - License compliance enforcement
- Supply chain security (detect suspicious sources) - Daily scheduled scans catch new vulnerabilities - Fail CI/CD on vulnerable dependencies

I-2: Limited Security Headers Configuration Guidance

Severity: Informational **Category:** Security Configuration **SOC 2 Relevance:** Partial (CC6.1)

Description

Vaultwarden relies on reverse proxy deployments (nginx, Caddy, Traefik) for TLS termination and security headers. While this is a valid architectural choice, the documentation could provide more comprehensive guidance on recommended security headers to ensure deployments follow security best practices.

Current State

Security headers are reverse proxy responsibility: - No built-in security headers in Vaultwarden application - Documentation provides basic nginx/Caddy examples - No comprehensive security header checklist

Recommended Headers Often Missing: - Content-Security-Policy (CSP) -
Permissions-Policy - X-Content-Type-Options: nosniff - X-Frame-Options: DENY
- Referrer-Policy: no-referrer - Strict-Transport-Security (HSTS)

Recommendations

Priority: Low

1. Comprehensive nginx configuration example:

```
# /etc/nginx/sites-available/vaultwarden
server {
    listen 443 ssl http2;
    server_name vault.example.com;

    # TLS Configuration
    ssl_certificate /etc/letsencrypt/live/vault.example.com/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/vault.example.com/privkey.pem;
    ssl_protocols TLSv1.3 TLSv1.2;
    ssl_ciphers 'ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-RSA-AES256-GCM-SHA384';
    ssl_prefer_server_ciphers on;

    # Security Headers
    add_header Strict-Transport-Security "max-age=31536000; includeSubDomains; preload";
    add_header X-Content-Type-Options "nosniff" always;
    add_header X-Frame-Options "DENY" always;
    add_header X-XSS-Protection "1; mode=block" always;
    add_header Referrer-Policy "no-referrer" always;
    add_header Permissions-Policy "geolocation=(), microphone=(), camera=()" always;

    # Content Security Policy
    add_header Content-Security-Policy "default-src 'self'; script-src 'self'; style-src

    # Proxy Configuration
    location / {
        proxy_pass http://127.0.0.1:8080;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    # WebSocket support
    location /notifications/hub {
        proxy_pass http://127.0.0.1:8080;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
    }

    # Client body size limit
```

```
client_max_body_size 525M;  
}
```

2. Caddy configuration with security headers:

```
vault.example.com {  
    # TLS configuration (automatic via Let's Encrypt)  
    tls {  
        protocols tls1.3 tls1.2  
    }  
  
    # Security headers  
    header {  
        Strict-Transport-Security "max-age=31536000; includeSubDomains; preload"  
        X-Content-Type-Options "nosniff"  
        X-Frame-Options "DENY"  
        X-XSS-Protection "1; mode=block"  
        Referrer-Policy "no-referrer"  
        Permissions-Policy "geolocation=(), microphone=(), camera=()"  
        Content-Security-Policy "default-src 'self'; script-src 'self'; style-src 'self'  
    }  
  
    # Reverse proxy  
    reverse_proxy localhost:8080 {  
        header_up X-Real-IP {remote_host}  
    }  
}
```

3. Security header validation script:


```
#!/bin/bash
# check-security-headers.sh

URL="${1:-https://vault.example.com}"

echo "Checking security headers for: $URL"
echo "===== "

HEADERS=(
    "Strict-Transport-Security"
    "X-Content-Type-Options"
    "X-Frame-Options"
    "Referrer-Policy"
    "Permissions-Policy"
    "Content-Security-Policy"
)

for HEADER in "${HEADERS[@]}; do
    VALUE=$(curl -sI "$URL" | grep -i "^${HEADER}:" | cut -d' ' -f2-)
    if [ -z "$VALUE" ]; then
        echo "❌ $HEADER: MISSING"
    else
        echo "✅ $HEADER: $VALUE"
    fi
done
```

4. Documentation additions:

Security Headers Checklist

Before deploying Vaultwarden to production, verify these security headers are configured

Required Headers

- [] `Strict-Transport-Security` - Force HTTPS for 1 year
- [] `X-Content-Type-Options: nosniff` - Prevent MIME sniffing
- [] `X-Frame-Options: DENY` - Prevent clickjacking
- [] `Referrer-Policy: no-referrer` - Prevent referrer leakage

Recommended Headers

- [] `Content-Security-Policy` - Mitigate XSS attacks
- [] `Permissions-Policy` - Restrict browser features
- [] `X-XSS-Protection` - Legacy XSS protection (for older browsers)

Validation

Run: ``curl -I https://your-vault.com | grep -E 'Strict-Transport|X-Frame|X-Content'``

Online tools:

- <https://securityheaders.com>
- <https://observatory.mozilla.org>

Benefits: - Improved deployment security out-of-the-box - Easier compliance with security standards - Better protection against XSS, clickjacking, and MITM attacks - Simplified security audit process

I-3: Database Backup Strategy Documentation Gap

Severity: Informational **Category:** Operational Security **SOC 2 Relevance:** Yes (CC7.3 - Data Retention)

Description

While Vaultwarden provides built-in database backup functionality for SQLite (via admin panel), there is limited documentation on comprehensive backup strategies for all supported databases (SQLite, MySQL, PostgreSQL), disaster recovery procedures, and backup security best practices.

Current State

SQLite Backup Present:

File: `src/api/admin.rs` - Backup functionality

```
#[cfg(sqlite)]
static CAN_BACKUP: LazyLock<bool> =
    LazyLock::new(|| ACTIVE_DB_TYPE.get().map(|t| *t == DbConnType::Sqlite).unwrap_or(f
```

Gaps: - No documented backup retention policies - No backup encryption guidance - Limited MySQL/PostgreSQL backup documentation - No disaster recovery testing procedures - No automated backup verification

Recommendations

Priority: Medium (for production deployments)

1. Comprehensive backup documentation:

```
# Vaultwarden Backup & Recovery Guide

## SQLite Backup

### Built-in Admin Panel Backup
1. Navigate to Admin Panel → Backup Database
2. Download `db.sqlite3.backup` file
3. Store securely with encryption

### Automated SQLite Backup Script
```bash
#!/bin/bash
backup-vaultwarden-sqlite.sh

BACKUP_DIR="/backup/vaultwarden"
DATA_DIR="/var/lib/vaultwarden"
TIMESTAMP=$(date +%Y%m%d_%H%M%S)
RETENTION_DAYS=30

Create backup directory
mkdir -p "$BACKUP_DIR"

Backup database with SQLite online backup
sqlite3 "$DATA_DIR/db.sqlite3" ".backup '$BACKUP_DIR/db_${TIMESTAMP}.sqlite3'"

Backup attachments and other data
tar -czf "$BACKUP_DIR/data_${TIMESTAMP}.tar.gz" \
 "$DATA_DIR/attachments" \
 "$DATA_DIR/sends" \
 "$DATA_DIR/icon_cache" \
 "$DATA_DIR/rsa_key.pem"

Encrypt backups
gpg --encrypt --recipient backup@example.com \
 "$BACKUP_DIR/db_${TIMESTAMP}.sqlite3"
gpg --encrypt --recipient backup@example.com \
 "$BACKUP_DIR/data_${TIMESTAMP}.tar.gz"

Remove unencrypted backups
rm "$BACKUP_DIR/db_${TIMESTAMP}.sqlite3"
```

```
rm "$BACKUP_DIR/data_${TIMESTAMP}.tar.gz"

Delete old backups
find "$BACKUP_DIR" -name "*.gpg" -mtime +$RETENTION_DAYS -delete

echo "Backup completed: $TIMESTAMP"
```

## MySQL/MariaDB Backup

```
#!/bin/bash
backup-vaultwarden-mysql.sh

MYSQL_USER="vaultwarden"
MYSQL_PASSWORD="your_password"
MYSQL_DATABASE="vaultwarden"
BACKUP_DIR="/backup/vaultwarden"
TIMESTAMP=$(date +%Y%m%d_%H%M%S)

mysqldump --user="$MYSQL_USER" \
 --password="$MYSQL_PASSWORD" \
 --single-transaction \
 --routines \
 --triggers \
 "$MYSQL_DATABASE" | gzip > "$BACKUP_DIR/mysql_${TIMESTAMP}.sql.gz"

Encrypt backup
gpg --encrypt --recipient backup@example.com \
 "$BACKUP_DIR/mysql_${TIMESTAMP}.sql.gz"
rm "$BACKUP_DIR/mysql_${TIMESTAMP}.sql.gz"
```

## PostgreSQL Backup

```
#!/bin/bash
backup-vaultwarden-postgres.sh

PGUSER="vaultwarden"
PGDATABASE="vaultwarden"
BACKUP_DIR="/backup/vaultwarden"
TIMESTAMP=$(date +%Y%m%d_%H%M%S)

pg_dump --username="$PGUSER" \
 --format=custom \
 --file="$BACKUP_DIR/postgres_${TIMESTAMP}.dump" \
 "$PGDATABASE"

Encrypt backup
gpg --encrypt --recipient backup@example.com \
 "$BACKUP_DIR/postgres_${TIMESTAMP}.dump"
rm "$BACKUP_DIR/postgres_${TIMESTAMP}.dump"
```

## Backup Verification

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### Automated Backup Testing

```
#!/bin/bash
verify-backup.sh

BACKUP_FILE="$1"
TEST_DIR="/tmp/backup-test"

Extract backup
mkdir -p "$TEST_DIR"
gpg --decrypt "$BACKUP_FILE" | tar -xzf - -C "$TEST_DIR"

Verify SQLite integrity
sqlite3 "$TEST_DIR/db.sqlite3" "PRAGMA integrity_check;"

if [$? -eq 0]; then
 echo "✅ Backup verified successfully"
 rm -rf "$TEST_DIR"
 exit 0
else
 echo "❌ Backup verification failed!"
 exit 1
fi
```

## Disaster Recovery Procedure

---

### SQLite Recovery

1. Stop Vaultwarden service: `systemctl stop vaultwarden`
2. Decrypt backup: `gpg --decrypt db_TIMESTAMP.sqlite3.gpg > db.sqlite3`
3. Replace database: `cp db.sqlite3 /var/lib/vaultwarden/`
4. Set permissions: `chown vaultwarden:vaultwarden /var/lib/vaultwarden/db.sqlite3`
5. Start service: `systemctl start vaultwarden`
6. Verify login and data integrity

## MySQL Recovery

1. Stop Vaultwarden service
2. Decrypt backup: `gpg --decrypt mysql_TIMESTAMP.sql.gz.gpg | gunzip > restore.sql`
3. Restore database: `mysql -u vaultwarden -p vaultwarden < restore.sql`
4. Start Vaultwarden service
5. Verify functionality

## PostgreSQL Recovery

1. Stop Vaultwarden service
2. Decrypt backup: `gpg --decrypt postgres_TIMESTAMP.dump.gpg > restore.dump`
3. Restore database: `pg_restore -U vaultwarden -d vaultwarden restore.dump`
4. Start Vaultwarden service
5. Verify functionality

## Backup Best Practices

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### Retention Policy

- **Daily backups:** Keep for 7 days
- **Weekly backups:** Keep for 4 weeks
- **Monthly backups:** Keep for 12 months
- **Yearly backups:** Keep for 7 years (compliance)

### Security

-  Always encrypt backups with GPG or similar
-  Store backups on separate infrastructure
-  Use offsite or cloud storage for redundancy
-  Test backup restoration quarterly
-  Monitor backup success/failure
-  Document recovery procedures
-  Restrict backup file access (0600 permissions)



## Automation

- Schedule backups via cron (daily at minimum)
- Send alerts on backup failure
- Verify backup integrity automatically
- Monitor backup storage capacity

## Example Cron Configuration

```
Daily backup at 2 AM
0 2 * * * /usr/local/bin/backup-vaultwarden.sh

Weekly backup verification on Sundays at 3 AM
0 3 * * 0 /usr/local/bin/verify-latest-backup.sh

Monthly offsite sync on 1st of month at 4 AM
0 4 1 * * rsync -avz /backup/vaultwarden/ backup-server:/vaultwarden/
```

## Monitoring

---

### Backup Monitoring Script

```
#!/bin/bash
monitor-backups.sh

BACKUP_DIR="/backup/vaultwarden"
MAX_AGE_HOURS=26 # Alert if no backup in 26 hours

LATEST_BACKUP=$(find "$BACKUP_DIR" -name "*.gpg" -type f -mtime -1 | head -n 1)

if [-z "$LATEST_BACKUP"]; then
 echo "❌ WARNING: No backup found in last 24 hours!"
 # Send alert (email, Slack, PagerDuty, etc.)
 exit 1
else
 echo "✅ Backup found: $LATEST_BACKUP"
 exit 0
fi
```

**\*\*2. Add backup status to admin panel:\*\***

```
```rust
// Display backup age and status in admin panel
#[get("/admin/backup-status")]
async fn backup_status(_token: AdminToken) -> JsonResult {
    let backup_dir = Path::new(&CONFIG.data_folder()).join("backups");

    let latest_backup = std::fs::read_dir(backup_dir)?
        .filter_map(|e| e.ok())
        .filter(|e| e.path().extension().map_or(false, |ext| ext == "sqlite3"))
        .max_by_key(|e| e.metadata().ok()?.modified().ok()?);

    if let Some(backup) = latest_backup {
        let age = SystemTime::now()
            .duration_since(backup.metadata()?.modified()?)?
            .as_secs() / 3600; // Hours

        Ok(Json(json!({
            "last_backup": backup.file_name(),
            "age_hours": age,
            "status": if age < 24 { "healthy" } else { "warning" }
        })))
    } else {
        Ok(Json(json!({
            "status": "no_backup",
            "message": "No backups found"
        })))
    }
}
```

Benefits: - Comprehensive disaster recovery capability - Reduced data loss risk - SOC 2 compliance for data retention - Documented recovery procedures - Automated backup verification

I-4: No Rate Limiting on Password Hint Requests

Severity: Informational **Category:** Denial of Service / Information Disclosure **SOC 2 Relevance:** Partial (CC6.1)

Description

Password hint retrieval endpoints (when enabled) lack aggressive rate limiting, potentially allowing attackers to enumerate user accounts or harvest password hints in bulk. While password hints are an optional feature, the endpoint should have stricter rate limiting than standard authentication endpoints.

Current State

Standard rate limiting applies: - General API rate limits apply - No hint-specific rate limiting - Potential for bulk hint harvesting

Recommendations

Priority: Low (if `PASSWORD_HINTS_ALLOWED=true`)

Add aggressive rate limiting for hint requests:

```
// src/api/core/accounts.rs

use crate::ratelimit::check_limit_login; // Reuse existing rate limiter

#[post("/accounts/password-hint", data = "<data>")]
async fn password_hint(data: Json<PasswordHintData>, ip: ClientIp, conn: DbConn) -> Emp

    if !CONFIG.password_hints_allowed() {
        err!("Password hints are disabled");
    }

    // Aggressive rate limiting: 3 requests per hour per IP
    if !rate_limit_hint_request(&ip.ip, 3, 3600) {
        err!("Too many password hint requests. Please try again later.", 429);
    }

    let data: PasswordHintData = data.into_inner();
    let email = &data.Email;

    // Log hint request for monitoring
    info!("Password hint requested for email: {} from IP: {}", email, ip.ip);

    match User::find_by_mail(email, &conn).await {
        Some(user) => {
            if let Some(hint) = user.password_hint {
                mail::send_password_hint(&user.email, hint).await?;
            }
            // Always return success to prevent user enumeration
            Ok(())
        }
        None => {
            // Constant-time response even for non-existent users
            tokio::time::sleep(Duration::from_millis(100)).await;
            Ok(())
        }
    }
}
```

Benefits: - Prevents bulk hint harvesting - Reduces user enumeration risk - Limits information disclosure

I-5: Limited Monitoring and Observability

Severity: Informational **Category:** Operations & Monitoring **SOC 2 Relevance:** Yes (CC7.2 - System Monitoring)

Description

Vaultwarden provides basic logging but lacks built-in metrics endpoints (Prometheus), health check endpoints, and structured logging for modern observability tools. This makes it challenging to monitor application health, performance, and security events in production environments.

Current State

Available: - Log output to stdout/syslog - Basic event logging to database - Admin panel diagnostics page

Missing: - Prometheus metrics endpoint - Structured JSON logging - Health/readiness endpoints - Performance metrics (request latency, DB query time) - Security event metrics (failed logins, 2FA attempts)

Recommendations

Priority: Medium (for production deployments)

1. Add Prometheus metrics endpoint:

```

// Cargo.toml
[dependencies]
prometheus = "0.13"
lazy_static = "1.4"

// src/api/metrics.rs
use prometheus::{Encoder, TextEncoder, Counter, Histogram, register_counter, register_h
use rocket::Route;

lazy_static! {
    static ref HTTP_REQUESTS_TOTAL: Counter = register_counter!(
        "vaultwarden_http_requests_total",
        "Total HTTP requests"
    ).unwrap();

    static ref LOGIN_ATTEMPTS_TOTAL: Counter = register_counter!(
        "vaultwarden_login_attempts_total",
        "Total login attempts"
    ).unwrap();

    static ref LOGIN_FAILURES_TOTAL: Counter = register_counter!(
        "vaultwarden_login_failures_total",
        "Failed login attempts"
    ).unwrap();

    static ref ACTIVE_WEBSOCKET_CONNECTIONS: Gauge = register_gauge!(
        "vaultwarden_websocket_connections",
        "Active WebSocket connections"
    ).unwrap();

    static ref DB_QUERY_DURATION: Histogram = register_histogram!(
        "vaultwarden_db_query_duration_seconds",
        "Database query duration"
    ).unwrap();
}

pub fn routes() -> Vec<Route> {
    routes![metrics, health, ready]
}

```

```
#[get("/metrics")]
fn metrics(_token: AdminToken) -> String {
    let encoder = TextEncoder::new();
    let metric_families = prometheus::gather();
    let mut buffer = vec![];
    encoder.encode(&metric_families, &mut buffer).unwrap();
    String::from_utf8(buffer).unwrap()
}

#[get("/health")]
fn health() -> &'static str {
    "OK"
}

#[get("/ready")]
async fn ready(conn: DbConn) -> Result<&'static str, Status> {
    // Check database connectivity
    match conn.run(|c| c.execute("SELECT 1")).await {
        Ok(_) => Ok("Ready"),
        Err(_) => Err(Status::ServiceUnavailable),
    }
}
```

2. Structured JSON logging:


```
// Add structured logging support
use serde_json::json;

fn log_security_event(event_type: &str, details: Value) {
    if CONFIG.json_logging_enabled() {
        let log_entry = json!({
            "timestamp": Utc::now().to_rfc3339(),
            "level": "INFO",
            "event_type": event_type,
            "details": details,
        });
        println!("{}", log_entry);
    } else {
        info!("Security Event: {} - {:?}", event_type, details);
    }
}

// Usage
log_security_event("login_attempt", json!({
    "email": email,
    "ip": ip.to_string(),
    "success": true,
    "2fa_required": true,
})));
```

3. Grafana dashboard configuration:

```
# grafana-dashboard.json (excerpt)
{
  "dashboard": {
    "title": "Vaultwarden Monitoring",
    "panels": [
      {
        "title": "Login Attempts",
        "targets": [
          {
            "expr": "rate(vaultwarden_login_attempts_total[5m])"
          }
        ]
      },
      {
        "title": "Failed Logins",
        "targets": [
          {
            "expr": "rate(vaultwarden_login_failures_total[5m])"
          }
        ]
      },
      {
        "title": "Active WebSocket Connections",
        "targets": [
          {
            "expr": "vaultwarden_websocket_connections"
          }
        ]
      },
      {
        "title": "Database Query Latency",
        "targets": [
          {
            "expr": "histogram_quantile(0.95, vaultwarden_db_query_duration_seconds_buc"
          }
        ]
      }
    ]
  }
}
```

Benefits: - Real-time performance monitoring - Security event visibility - Capacity planning data
- Incident response capabilities - Integration with modern observability platforms

I-6: Session Management Documentation Enhancement

Severity: Informational **Category:** Security Configuration **SOC 2 Relevance:** Yes (CC6.1)

Description

While Vaultwarden implements secure session management (JWT tokens with 2-hour expiry, refresh tokens, security stamps), the documentation could provide more comprehensive guidance on session configuration, timeout policies, and session invalidation strategies for different deployment scenarios.

Recommendations

Priority: Low

Enhanced session management documentation:

```
# Vaultwarden Session Management Guide

## Session Architecture

Vaultwarden uses a multi-layered session management approach:

1. **Access Tokens (JWT)**: 2-hour expiry, contains user claims and security stamp
2. **Refresh Tokens**: 30-90 day expiry based on device trust level
3. **Security Stamps**: Invalidate all sessions on password change
4. **Device Binding**: Track trusted devices with extended refresh token lifetime

## Configuration Options

### Access Token Lifetime
```bash
Not configurable (hardcoded to 2 hours)
This is intentional for security - use refresh tokens for longer sessions
```

## Admin Session Lifetime

```
.env
ADMIN_SESSION_LIFETIME=20 # Minutes, default: 20
```

## Refresh Token Lifetime

- **Trusted devices** (remember me): 90 days
- **Untrusted devices**: 30 days

## Session Invalidation Strategies

---

### 1. Password Change → Invalidate All Sessions

```
// Automatically invalidates all sessions via security stamp rotation
user.reset_security_stamp();
user.save(&conn).await?;
```

### 2. Manual Session Revocation

Navigate to Admin Panel → Users → Deauth User

### 3. Device-specific Revocation

Users can remove devices from Account Settings → Devices

### 4. Organization-wide Session Reset

```
Emergency: Invalidate all user sessions
UPDATE users SET security_stamp = uuid();
```

## Security Best Practices

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### 1. Enforce Regular Re-authentication

For high-security environments, consider: - Shorter admin session lifetime (10 minutes) - Disable "Remember Me" option - Require 2FA on every login

## 2. Monitor Session Activity

- Review device list regularly
- Check for suspicious IP addresses
- Monitor failed login attempts

## 3. Session Security Checklist

- ☐ Admin session timeout < 30 minutes
- ☐ Users review connected devices monthly
- ☐ Failed login alerts configured
- ☐ Security stamp exceptions monitored
- ☐ Device trust policy documented

## Compliance Considerations (SOC 2)

---

### CC6.1 - Logical Access

- ☒ Session timeout enforced (2-hour access token)
- ☒ Device binding implemented
- ☒ Session revocation on password change
-  Consider: Idle timeout detection (client-side)

### CC6.7 - Removal of Access

- ☒ Immediate session invalidation via security stamp
- ☒ Device revocation capability
- ☒ Admin can force user logout

---

### ### I-7: Container Security Hardening Opportunities

**\*\*Severity:\*\*** Informational  
**\*\*Category:\*\*** Deployment Security  
**\*\*SOC 2 Relevance:\*\*** Partial (CC6.1)

#### #### Description

While Vaultwarden provides Docker images, there are opportunities to enhance container

#### #### Recommendations

**\*\*Priority:** Low to Medium (for containerized deployments)\*\*

**\*\*1. Enhanced Dockerfile security:\*\***

```
```dockerfile
# Dockerfile.hardened
FROM rust:1.90-slim as builder

# Run as non-root during build
RUN groupadd -r vaultwarden && useradd -r -g vaultwarden vaultwarden

# Security: Use specific package versions and verify checksums
RUN apt-get update && apt-get install -y --no-install-recommends \
    build-essential=12.9 \
    libssl-dev=3.0.15-1 \
    pkg-config=1.8.1-1 \
    && rm -rf /var/lib/apt/lists/*

WORKDIR /build
COPY --chown=vaultwarden:vaultwarden . .

USER vaultwarden
RUN cargo build --release --locked

FROM debian:bookworm-slim
```

```
# Security: Install only runtime dependencies
RUN apt-get update && apt-get install -y --no-install-recommends \
    ca-certificates=20230311 \
    libssl3=3.0.15-1 \
    && rm -rf /var/lib/apt/lists/*

# Create non-root user
RUN groupadd -r -g 999 vaultwarden && \
    useradd -r -u 999 -g vaultwarden vaultwarden && \
    mkdir -p /data /templates && \
    chown -R vaultwarden:vaultwarden /data /templates

# Security: Read-only filesystem except data directory
COPY --from=builder --chown=vaultwarden:vaultwarden /build/target/release/vaultwarden /

USER vaultwarden
WORKDIR /data

# Security: Drop all capabilities
VOLUME /data
EXPOSE 8080
ENTRYPOINT ["/usr/local/bin/vaultwarden"]

# Security labels
LABEL security.hardened="true" \
    security.user="vaultwarden:999" \
    security.capabilities="none"
```

2. Docker Compose with security options:

```
# docker-compose.hardened.yml
version: '3'

services:
  vaultwarden:
    image: vaultwarden/server:latest
    container_name: vaultwarden

    # Security: Read-only root filesystem
    read_only: true

    # Security: Tmpfs for temporary files
    tmpfs:
      - /tmp:noexec,nosuid,nodev,size=100m

    # Security: Drop all capabilities
    cap_drop:
      - ALL

    # Security: Add only required capabilities (if needed)
    # cap_add:
    #   - NET_BIND_SERVICE

    # Security: No new privileges
    security_opt:
      - no-new-privileges:true

    # Security: AppArmor profile (if available)
    # security_opt:
    #   - apparmor=docker-default

    # Security: Seccomp profile
    security_opt:
      - seccomp=./seccomp-profile.json

    # Resource limits
    mem_limit: 1g
    cpus: 1

    # Restart policy
```



```
restart: unless-stopped

# Network isolation
networks:
  - vaultwarden-net

volumes:
  - ./data:/data:rw
  - /etc/localtime:/etc/localtime:ro

environment:
  - DOMAIN=https://vault.example.com
  - SIGNUPS_ALLOWED=false
  - INVITATIONS_ALLOWED=true
  - ADMIN_TOKEN=${ADMIN_TOKEN}
  - LOG_LEVEL=info

# Health check
healthcheck:
  test: ["CMD", "wget", "-q", "--spider", "http://localhost:8080/alive"]
  interval: 30s
  timeout: 3s
  retries: 3
  start_period: 10s

networks:
  vaultwarden-net:
    driver: bridge
    ipam:
      config:
        - subnet: 172.20.0.0/24
```

3. Kubernetes security context:

```
# vaultwarden-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: vaultwarden
spec:
  replicas: 1
  selector:
    matchLabels:
      app: vaultwarden
  template:
    metadata:
      labels:
        app: vaultwarden
    spec:
      # Security: Run as non-root
      securityContext:
        runAsNonRoot: true
        runAsUser: 999
        runAsGroup: 999
        fsGroup: 999
        seccompProfile:
          type: RuntimeDefault

      containers:
        - name: vaultwarden
          image: vaultwarden/server:latest

      # Security: Container-level restrictions
      securityContext:
        allowPrivilegeEscalation: false
        readOnlyRootFilesystem: true
        capabilities:
          drop:
            - ALL

      # Resource limits
      resources:
        requests:
          memory: "256Mi"
```

```
    cpu: "250m"
  limits:
    memory: "1Gi"
    cpu: "1000m"

# Volumes
volumeMounts:
- name: data
  mountPath: /data
- name: tmp
  mountPath: /tmp

# Health checks
livenessProbe:
  httpGet:
    path: /alive
    port: 8080
  initialDelaySeconds: 30
  periodSeconds: 10

readinessProbe:
  httpGet:
    path: /alive
    port: 8080
  initialDelaySeconds: 5
  periodSeconds: 5

env:
- name: DOMAIN
  value: "https://vault.example.com"
- name: ADMIN_TOKEN
  valueFrom:
    secretKeyRef:
      name: vaultwarden-secrets
      key: admin-token

volumes:
- name: data
  persistentVolumeClaim:
    claimName: vaultwarden-data
```

```
- name: tmp
  emptyDir: {}

# Pod security policy
automountServiceAccountToken: false
```

Benefits: - Defense in depth for container deployments - Reduced attack surface - Compliance with CIS Docker/Kubernetes benchmarks - Better resource isolation

I-8: Security Testing and Continuous Validation

Severity: Informational **Category:** Security Assurance **SOC 2 Relevance:** Yes (CC7.1)

Description

While Vaultwarden has good code quality practices (clippy, rustfmt, forbidden unsafe code), there are opportunities to enhance security testing through fuzzing, static analysis security testing (SAST), and continuous security validation in CI/CD.

Recommendations

Priority: Medium

1. Add fuzzing for critical parsers:

```
// fuzz/fuzz_targets/fuzz_jwt.rs
#![no_main]
use libfuzzer_sys::fuzz_target;
use vaultwarden::auth::decode_jwt;

fuzz_target!(|data: &[u8]| {
    if let Ok(s) = std::str::from_utf8(data) {
        let _ = decode_jwt(s);
    }
});
```

```
# .github/workflows/fuzz.yml
name: Continuous Fuzzing

on:
  schedule:
    - cron: '0 0 * * 0' # Weekly

jobs:
  fuzz:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Install cargo-fuzz
        run: cargo install cargo-fuzz

      - name: Run fuzzing tests
        run: |
          cargo fuzz run fuzz_jwt -- -max_total_time=3600
          cargo fuzz run fuzz_api_input -- -max_total_time=3600
```

2. Enhanced SAST with cargo-geiger:

```
# .github/workflows/sast.yml
name: Security Analysis

on: [push, pull_request]

jobs:
  cargo-geiger:
    name: Unsafe Code Detection
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Install cargo-geiger
        run: cargo install cargo-geiger

      - name: Scan for unsafe code
        run: cargo geiger --all-features --output-format Json > geiger-report.json

      - name: Upload results
        uses: actions/upload-artifact@v4
        with:
          name: geiger-report
          path: geiger-report.json
```

3. Dependency license scanning:

```
# .github/workflows/license-check.yml
name: License Compliance

on: [push, pull_request]

jobs:
  license-check:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Install cargo-license
        run: cargo install cargo-license

      - name: Check licenses
        run: cargo license --all-features --json > licenses.json

      - name: Validate approved licenses
        run: |
          # Fail if GPL-3.0 or other copyleft licenses detected
          if grep -q "GPL-3.0" licenses.json; then
            echo "ERROR: GPL-3.0 license detected!"
            exit 1
          fi
```

4. Continuous penetration testing:

```
# .github/workflows/pentest.yml
name: Security Testing

on:
  schedule:
    - cron: '0 2 * * 1' # Weekly on Monday

jobs:
  zap-scan:
    name: OWASP ZAP Scan
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Start Vaultwarden
        run: docker-compose up -d

      - name: Wait for service
        run: sleep 30

      - name: Run ZAP baseline scan
        uses: zaproxy/action-baseline@v0.12.0
        with:
          target: 'http://localhost:8080'
          rules_file_name: '.zap/rules.tsv'
          cmd_options: '-a'
```

Benefits: - Continuous security validation - Early vulnerability detection - License compliance assurance - Reduced security debt

Part II: Comprehensive Security Recommendations

Priority Implementation Matrix

Immediate Actions (Week 1-2)

Priority	Finding	Action	Effort	Impact
Critical	M-1	Add startup warning for plain-text ADMIN_TOKEN	2 hours	High
Critical	M-3	Disable PASSWORD_HINTS_ALLOWED by default	1 hour	High
High	L-1	Implement admin action audit logging	8 hours	High
High	I-1	Add cargo-audit to CI/CD	4 hours	Medium
Medium	L-2	Add WebSocket rate limiting	6 hours	Medium

Short-term Actions (Month 1-3)

Priority	Finding	Action	Effort	Impact
High	M-1	Provide admin token hashing utility	8 hours	High
High	M-2	Simplify security stamp exception logic	16 hours	Medium
High	L-4	Implement CSV event log export	8 hours	Medium
Medium	L-3	Add admin panel IP allowlisting	6 hours	Medium
Medium	I-3	Enhance backup documentation	4 hours	Medium
Medium	I-5	Add Prometheus metrics	12 hours	Medium

Long-term Actions (Month 6-12)

Priority	Finding	Action	Effort	Impact
High	M-1	Deprecate plain-text ADMIN_TOKEN (breaking)	16 hours	High
Medium	M-3	Deprecate password hints feature	12 hours	Medium
Medium	L-3	Implement admin panel MFA	20 hours	High
Low	I-2	Enhanced security headers docs	4 hours	Low
Low	I-8	Continuous security testing	16 hours	Medium

SOC 2 Compliance Roadmap

Trust Services Criteria Coverage

CC6.1 - Logical and Physical Access Controls

Current State: Good -  Strong authentication (JWT, 2FA, WebAuthn) -  Role-based access control -  Plain-text admin tokens accepted -  Password hints in plaintext

Required Actions: 1. Enforce Argon2id ADMIN_TOKEN only 2. Disable or encrypt password hints 3. Add admin panel MFA option 4. Document access control policies

Timeline: 3 months

CC6.7 - Removal of Access

Current State: Good -  Security stamp invalidation -  Device revocation -  Admin can force logout

Required Actions: 1. Document session management procedures 2. Enhance admin panel user management logging

Timeline: 1 month

CC7.1 - System Operations (Vulnerability Management)

Current State: Moderate -  Trivy container scanning -  No automated Rust dependency scanning -  No continuous security testing

Required Actions: 1. Add cargo-audit to CI/CD 2. Implement cargo-deny policy 3. Schedule regular security scans 4. Document vulnerability response process

Timeline: 2 months

CC7.2 - System Monitoring

Current State: Moderate -  Event logging system -  Limited admin action logging -  No metrics/observability

Required Actions: 1. Enhance admin action logging 2. Implement Prometheus metrics 3. Add health check endpoints 4. Document monitoring procedures

Timeline: 3 months

CC7.3 - Data Retention

Current State: Basic -  Event retention configurable -  SQLite backup functionality -  Limited backup documentation

Required Actions: 1. Document comprehensive backup strategy 2. Implement automated backup testing 3. Define retention policies 4. Document disaster recovery procedures

Timeline: 1 month

Security Testing Strategy

1. Static Analysis

Tools: - `cargo clippy --all-features -- -D warnings` - `cargo audit` - Dependency vulnerabilities - `cargo deny check` - License and policy compliance - `cargo geiger` - Unsafe code detection

Frequency: Every commit (CI/CD)

2. Dynamic Analysis

Tools: - OWASP ZAP - Web application scanning - Burp Suite - Manual penetration testing - sqlmap - SQL injection testing (should fail) - nuclei - Template-based vulnerability scanning

Frequency: - Weekly automated (ZAP baseline) - Monthly manual testing - Quarterly third-party penetration test

3. Fuzzing

Targets: - JWT parsing - JSON API input - Email address validation - Organization/collection name handling

Tools: - cargo-fuzz with libFuzzer - AFL++ (American Fuzzy Lop)

Frequency: Continuous (dedicated server) or weekly

4. Dependency Scanning

Tools: - cargo-audit (RustSec Advisory Database) - Trivy (container image scanning) - Dependabot (GitHub automated PRs)

Frequency: Daily automated scans

5. Code Review

Focus Areas: - Authentication logic changes - Authorization checks - Cryptographic implementations - Database queries - File operations - External API calls

Process: - Mandatory peer review for all PRs - Security-focused review for sensitive changes - External security audit annually

```
# Firewall rules (iptables example)
# Allow only HTTPS and SSH
iptables -A INPUT -p tcp --dport 443 -j ACCEPT
iptables -A INPUT -p tcp --dport 22 -s 10.0.0.0/8 -j ACCEPT
iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
iptables -P INPUT DROP
iptables -P FORWARD DROP
```

```
# Strong TLS configuration
ssl_protocols TLSv1.3 TLSv1.2;
ssl_ciphers 'ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-RSA-AES256-GCM-SHA384:ECDHE-ECDSA-CHACHA20-POLY1305:ECDHE-RSA-CHACHA20-POLY1305:ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-RSA-AES128-GCM-SHA256:ECDHE-ECDSA-AES256-SHA384:ECDHE-RSA-AES256-SHA384:ECDHE-ECDSA-AES128-SHA256:ECDHE-RSA-AES128-SHA256:ECDHE-ECDSA-AES256-SHA:ECDHE-RSA-AES256-SHA:ECDHE-ECDSA-AES128-SHA:ECDHE-RSA-AES128-SHA:ECDHE-ECDSA-DES-CBC3-SHA:ECDHE-RSA-DES-CBC3-SHA:ECDHE-ECDSA-DES-CBC3-SHA:ECDHE-RSA-DES-CBC3-SHA:ECDHE-ECDSA-DES-CBC3-SHA:ECDHE-RSA-DES-CBC3-SHA:ECDHE-ECDSA-DES-CBC3-SHA:ECDHE-RSA-DES-CBC3-SHA';
ssl_prefer_server_ciphers off;
ssl_session_cache shared:SSL:10m;
ssl_session_timeout 10m;
ssl_stapling on;
ssl_stapling_verify on;
```

3. Operating System Hardening

```
# File permissions
chmod 600 /var/lib/vaultwarden/.env
chmod 600 /var/lib/vaultwarden/db.sqlite3
chown -R vaultwarden:vaultwarden /var/lib/vaultwarden

# AppArmor profile (Ubuntu/Debian)
cat > /etc/apparmor.d/usr.local.bin.vaultwarden <<EOF
#include <tunables/global>

/usr/local/bin/vaultwarden {
    #include <abstractions/base>
    #include <abstractions/nameservice>
    #include <abstractions/openssl>

    /usr/local/bin/vaultwarden mr,
    /var/lib/vaultwarden/** rwk,
    /tmp/** rwk,

    # Deny dangerous operations
    deny /proc/** w,
    deny /sys/** w,
    deny /dev/** w,
}
EOF

apparmor_parser -r /etc/apparmor.d/usr.local.bin.vaultwarden
```

4. Vaultwarden Configuration Hardening

```
# .env - Security-hardened configuration

# Disable signups (invitation-only)
SIGNUPS_ALLOWED=false
INVITATIONS_ALLOWED=true

# Email domain whitelist
SIGNUPS_DOMAINS_WHITELIST=example.com,company.org

# Disable password hints
PASSWORD_HINTS_ALLOWED=false

# Strong admin token (Argon2id hashed)
ADMIN_TOKEN='$argon2id$v=19$m=65540,t=3,p=4$...'

# Short admin session
ADMIN_SESSION_LIFETIME=10

# Enable org events
ORG_EVENTS_ENABLED=true

# Disable web vault if using clients only
WEB_VAULT_ENABLED=true

# Rate limiting (aggressive)
LOGIN_RATELIMIT_SECONDS=60
LOGIN_RATELIMIT_MAX_BURST=3

# SMTP for notifications only (no hint emails)
SMTP_HOST=smtp.example.com
SMTP_FROM=vaultwarden@example.com
SMTP_SECURITY=starttls

# Logging
LOG_LEVEL=info
EXTENDED_LOGGING=true
USE_SYSLOG=true

# Database
DATABASE_URL=/data/db.sqlite3
```



```
ENABLE_DB_WAL=true
```

```
# Icon service security
```

```
ICON_SERVICE=internal
```

```
ICON_CACHE_TTL=2592000
```

5. Monitoring and Alerting

```
# Prometheus alerting rules
groups:
- name: vaultwarden_security
  interval: 30s
  rules:
    - alert: HighFailedLoginRate
      expr: rate(vaultwarden_login_failures_total[5m]) > 0.5
      for: 5m
      labels:
        severity: warning
      annotations:
        summary: "High failed login rate detected"
        description: "{{ $value }}" failed logins per second over 5 minutes"

    - alert: NoRecentBackup
      expr: time() - vaultwarden_last_backup_timestamp > 86400
      for: 1h
      labels:
        severity: critical
      annotations:
        summary: "No backup in 24 hours"
        description: "Last backup was {{ $value }}" seconds ago"

    - alert: UnusualWebSocketConnections
      expr: vaultwarden_websocket_connections > 1000
      for: 10m
      labels:
        severity: warning
      annotations:
        summary: "Unusually high WebSocket connection count"
        description: "{{ $value }}" active connections"
```

Security Checklist

Pre-deployment Security Checklist

- ☐ **Authentication**
 - ☐ ADMIN_TOKEN is Argon2id hashed
 - ☐ PASSWORD_HINTS_ALLOWED is false
 - ☐ SIGNUPS_ALLOWED is false (or email whitelist configured)
 - ☐ 2FA enforced for admin accounts
- ☐ **Network Security**
 - ☐ TLS 1.3/1.2 configured
 - ☐ Security headers configured (HSTS, CSP, etc.)
 - ☐ Firewall rules restrict access
 - ☐ Reverse proxy properly configured
- ☐ **Data Protection**
 - ☐ Database file permissions 0600
 - ☐ Configuration file permissions 0600
 - ☐ Automated encrypted backups configured
 - ☐ Backup testing performed
- ☐ **Monitoring**
 - ☐ Logging enabled (syslog or file)
 - ☐ Failed login alerting configured
 - ☐ Backup monitoring configured
 - ☐ Health checks implemented
- ☐ **Updates**
 - ☐ Automated security updates enabled (OS)
 - ☐ Vaultwarden update process documented
 - ☐ Dependency scanning in CI/CD
 - ☐ Vulnerability response process documented

- [] **Documentation**
 - [] Disaster recovery procedures documented
 - [] Access control policies documented
 - [] Incident response plan created
 - [] Security contact information published
-

Incident Response Plan

1. Potential Security Incident

Indicators: - Unusually high failed login attempts - Unexpected admin panel access - Database backup missing or corrupted - Suspicious user account activity

Response: 1. **Assess:** Determine scope and severity 2. **Contain:** Block suspicious IPs, disable affected accounts 3. **Investigate:** Review logs, database, file integrity 4. **Remediate:** Apply fixes, rotate credentials 5. **Document:** Record timeline, actions, lessons learned

2. Compromised Admin Token

Actions: 1. Immediately change ADMIN_TOKEN 2. Restart Vaultwarden service 3. Review admin panel audit logs 4. Check for unauthorized changes 5. Review all user accounts for tampering 6. Notify users if data affected 7. Post-incident review

3. Database Compromise

Actions: 1. Take server offline immediately 2. Preserve evidence (disk image, logs) 3. Restore from last known good backup 4. Force password reset for all users (security stamp rotation) 5. Review breach scope 6. Notify affected users per data breach laws 7. Implement additional monitoring 8. Conduct security audit

4. Dependency Vulnerability

Actions: 1. Assess vulnerability severity (CVSS score) 2. Check if vulnerability affects Vaultwarden (reachable code) 3. Update dependency to patched version 4. Run full test suite 5. Deploy update urgently if critical 6. Document in changelog 7. Notify users if exploitation detected

Conclusion

Vaultwarden demonstrates a **strong security foundation** with excellent architectural choices:

Key Strengths

-  Memory-safe Rust implementation with forbidden unsafe code
-  Zero-knowledge architecture preserving user privacy
-  Strong cryptographic implementations (Argon2id, PBKDF2)
-  Comprehensive SSRF protection
-  Proper ORM usage preventing SQL injection
-  Multiple authentication factors supported

Areas for Enhancement

1. **Configuration Security:** Deprecate plain-text admin tokens
2. **Audit Logging:** Expand admin action logging
3. **Observability:** Add metrics and monitoring endpoints
4. **Documentation:** Enhance operational security guides
5. **Testing:** Implement continuous security validation

Final Recommendations

For Production Deployments: 1. Implement all **High Priority** findings within 3 months 2. Establish comprehensive backup and monitoring 3. Document incident response procedures 4. Perform quarterly security reviews 5. Keep dependencies updated with automated scanning

For SOC 2 Compliance: 1. Address all findings flagged "SOC 2 Relevance: Yes" 2. Implement comprehensive audit logging 3. Document security policies and procedures 4. Establish vulnerability management process 5. Conduct annual third-party security audit

Overall Risk Assessment

Current Risk Level: LOW to MEDIUM

Vaultwarden is suitable for production use with appropriate deployment hardening. The identified findings represent defense-in-depth improvements rather than critical vulnerabilities. With implementation of recommended Medium and High priority findings, the risk level reduces to **LOW**.

End of Security Audit Report Series

Report Summary: - **Report 1:** Executive Summary & Threat Model (18 pages) - **Report 2:** Critical & High Severity Findings (16 pages) - **Report 3:** Medium & Low Severity Findings (45 pages) - **Report 4:** Informational Findings & Recommendations (52 pages)

Total Findings: 0 Critical, 0 High, 3 Medium, 5 Low, 8 Informational

Audit Completion Date: January 2026

Appendix: Additional Resources

Security Tools

- **cargo-audit:** <https://github.com/rustsec/rustsec/tree/main/cargo-audit>
- **cargo-deny:** <https://github.com/EmbarkStudios/cargo-deny>
- **OWASP ZAP:** <https://www.zaproxy.org/>
- **Trivy:** <https://github.com/aquasecurity/trivy>

Security Standards

- **OWASP Top 10:** <https://owasp.org/www-project-top-ten/>
- **CWE Top 25:** <https://cwe.mitre.org/top25/>
- **SOC 2:** <https://www.aicpa.org/soc>

Documentation

- **Vaultwarden Wiki:** <https://github.com/dani-garcia/vaultwarden/wiki>
- **Rust Security:** <https://anssi-fr.github.io/rust-guide/>
- **Diesel Security:** <https://diesel.rs/guides/all-about-inserts.html>